A Pointwise Digit Certificate and a Cyclic-Modulus Counterexample

Qichao Wang Daoyu Dong

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Write $W_j = 1\|\cdots\|j$, $D(j) = \text{digits}(W_j)$, and $W_0 = D(0) = 0$. Moving the first k whole integer blocks to the end gives

$$R_{m,k} = 10^{D(k)}W_m - (10^{D(m)} - 1)W_k, \qquad 0 \le k < m. \tag{1}$$

The previous bounds $O(X)$ for square pairs and $O(X^2/\log X)$ for perfect-power pairs permit isolated exceptions. No counting bound is used below. All integer positions in (1), including the middle blocks, retain their prescribed values.

1 A complete cyclic modulus does not suffice

Proposition 1. *The actual rotation*

$$R = R_{16,1} = 23456789101112131415161, \qquad \Delta = 10^{23} - 1,$$

is a square in $\mathbb{Q}_p$ for every prime $p \mid 10\Delta$, but is not an integer square. Consequently it is a square modulo $10^a\Delta^b$ for every pair of nonnegative integers a, b. Thus even the complete cyclic modulus together with arbitrarily precise decimal congruences cannot reject every individual rotation.

Proof. The word is $2\|3\|\cdots\|16\|1$, with 23 digits, and (1) reads $R = 10W_{16} - \Delta$. Direct integer arithmetic gives

$$r = 24515341518503348803133, \qquad r^2 \equiv R \pmod{\Delta}, \qquad \gcd(2r, \Delta) = 1.$$

For every prime divisor p of Δ, this is a nonsingular root modulo p, which lifts to all p-adic precisions by Hensel's lemma. No factorization of Δ is needed. Also $R \equiv 1 \pmod 8$ and $R \equiv 1 \pmod 5$, giving square roots in $\mathbb{Q}_2$ and $\mathbb{Q}_5$. These are integral local roots. The Chinese remainder theorem proves the assertion for $10^a\Delta^b$. Section 3 proves that R is not an integer square using an actual middle digit. □

This is a counterexample to the sufficiency of these local conditions, not to Kashihara's conjecture. In particular, retaining all middle digits in a congruence is different from forcing a root of the required real size. No claim that every pointwise method fails follows.

2 Exact compatibility with the real interval

Let a positive integer R have E digits. Prescribe its first h digits P and its last q digits $U = R \bmod 10^q$, with $q \ge 3$. The digits of U are taken from the actual word, and may extend past the original moved prefix into its middle blocks. Put

$$A = \left\lceil \sqrt{P10^{E-h}} \right\rceil, \qquad B = \left\lfloor \sqrt{(P+1)10^{E-h} - 1} \right\rfloor.$$

Assume $v_2(U), v_5(U) < q$. An odd valuation excludes squares immediately. Otherwise write $v_2(U) = 2a$, $v_5(U) = 2b$, and

$$z = 2^a5^b, \qquad H = 10^q/z.$$

Solving the two unit congruences at 2 and 5 and applying CRT gives a complete set $\mathcal{A} \subset [0, H)$ of at most eight root classes:

$$y^2 \equiv U \pmod{10^q} \iff y \equiv r \pmod{H} \text{ for some } r \in \mathcal{A}. \tag{2}$$

Indeed, division by p^{2a} or p^{2b} gives unit square roots, at most four at 2 and two at 5. Multiplying back gives root moduli 2^{q-a} and 5^{q-b}, whose product is H.

Lemma 1 (Pointwise interval test)**.** *A square with the prescribed length, prefix and suffix exists if and only if, for some $r \in \mathcal{A}$,*

$$\max\left(0, \left\lceil \frac{A-r}{H} \right\rceil\right) \;\le\; \left\lfloor \frac{B-r}{H} \right\rfloor.$$

If $B - A < H$, at most eight possible positive roots occur. Failure of all these inequalities excludes the particular rotation supplying the digits.

Proof. The prefix and length are equivalent to $A \le y \le B$; the suffix is equivalent to (2). Write $y = r + jH$, with $j \ge 0$, and solve the two interval inequalities for j. An interval shorter than H contains at most one integer from each root class. □

The test is exact for the selected windows. If it leaves roots, the additional actual middle digits must be checked; their values cannot be filled freely. Increasing q preserves every genuine square candidate. At $q = E$, the test is an exact square decision, since $y^2 < 10^E$ and $y^2 \equiv R \pmod{10^E}$ force equality. This termination observation does not prove uniform exclusion for the infinite rotation family.

3 One middle digit excludes the isolated candidate

For Proposition 1, the first digit is 2, so

$$[A, B] = [141421356238,\ 173205080756].$$

At precision $q = 12$, the actual suffix is

$$U = 112131415161 = 1\|12\|13\|14\|15\|16\|1.$$

Its first displayed 1 is the final digit of the middle block 11; it is not part of the original moved prefix $W_1 = 1$. Here $H = 10^{12}$. The complete eight roots modulo H are

55164164619	104480570869	395519429131	444835835381
555164164619	604480570869	895519429131	944835835381.

Each entry squares to U modulo 10^{12}. They are distinct, and the unit-root bound of eight proves completeness. None belongs to $[A, B]$; also $B < H$, so no positive translate of a listed root can belong to the interval. Lemma 1 proves that R is not a square.

The extra middle digit is substantive. The weaker 11-digit suffix 12131415161 and the same first digit and total length admit the following four squares:

Root	Square	Twelfth digit from the right
144835835381	20977419210512131415161	5
145519429131	21175904254612131415161	6
154480570869	23864246776012131415161	0
155164164619	24075917981912131415161	9

All four have 23 digits, start with 2, and share the actual last 11 digits. The next actual digit is 1, whereas these completions have $5, 6, 0, 9$. The complete 12-digit calculation above excludes every completion, not merely these four examples.

4 Verification and the remaining problem

`verify_pointwise_middle.py` checks every displayed certificate exactly and compares `pointwise_rotation.cpp` (GMP/OpenMP) with the earlier Python root solver and small exhaustive intervals. The window and Hensel/CRT tools are standard; no priority claim is made. No uniform theorem proves that the interval test rejects every rotation, and square tests do not exclude odd prime powers. Kashihara's conjecture remains open; the existing sparsity bounds have not been strengthened.